

On Threshold Selection Using Clustering Criteria

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Abstract—The threshold selection method of Otsu is shown to break down for a certain range of object-to-background pixel population ratios. Modifications to Otsu's method are proposed to overcome some of its limitations. The findings are also relevant to the closely related methods of Ridler and Trussel.

INTRODUCTION

In our comparative study of threshold selection algorithms [1], the method advocated by Otsu [2], despite its simplicity, performed consistently well on the range of applications considered. However, further experiments with the method have revealed that it breaks down for certain ratios of populations of object and background pixels in an image which in practice may arise quite frequently.

The purpose of this correspondence is to explain the behavior of Otsu's threshold selection function under the condition of highly unequal population sizes. Our findings contradict Otsu's conjecture of unimodality for this function, and they have implications for the closely related threshold selection method of Ridler [3]. Modifications to these threshold selection methods will be proposed to overcome some of their current limitations.

OTSU METHOD: SHORTCOMINGS AND SUGGESTED CORRECTION

In the automatic thresholding method of Otsu the optimal threshold τ is selected by maximizing a measure $J_0(T)$ of separability of the gray-level populations obtained by partitioning the range of image gray-level values at the point T . The function $J_0(T)$ is defined as

$$J_0(T) = N_1(T)N_2(T)[\mu_1(T) - \mu_2(T)]^2 / [N_1(T) + N_2(T)] \quad (1)$$

where $\mu_i(T)$, $i = 1, 2$, is the mean of the i th population and $N_i(T)$ is the number of pixels constituting it.

Otsu observed in experimental studies that the function $J_0(T)$ always behaved well; namely, in each case it had just one maximum that corresponded to a good threshold value. These experimental results were used as supporting evidence to conjecture that $J_0(T)$ is unimodal irrespective of the gray level distribution. We shall demonstrate that this conjecture does not hold in general. The objective function may not only be multimodal, but more importantly, if it is multimodal, its global maximum is not guaranteed to give a correct threshold. Such a situation may arise for a certain range of the relative sizes of the populations of object and background pixels. More specifically, as the population sizes of background and object become greatly disparate, the criterion function becomes bimodal, with the second mode centered near the mean gray level of the larger population. The exact range of population size ratios over which $J_0(T)$ is bimodal will depend on the actual parameters (means and variances) of the two populations. With increasing population imbalance, the second mode grows in size while the first mode diminishes, until at very large disparities the objective function once more appears unimodal. A simple heuristic has been devised to recognize these

distinct situations, and this allows an assessment of the appropriateness of candidate thresholds to be made.

For example, consider a 512×512 image of a bright square of 50×50 pixels on the dark background shown in Fig. 1. The object and background pixel populations are normally distributed with a contrast of 80 gray levels and standard deviation of ten gray levels. The image histogram and objective function $J_0(T)$ are given in Figs. 2 and 3, respectively. The threshold corresponding to the global peak of $J_0(T)$ splits the large population of background pixels in half. The correct value of threshold is given by the local maximum at gray level $\tau = 134$. The magnitude of the peak is only 82 percent of the global maximum.

This finding affects the method of Otsu in two ways. First of all, the potential presence of two modes in the objective function means that it is not sufficient to perform only the relatively

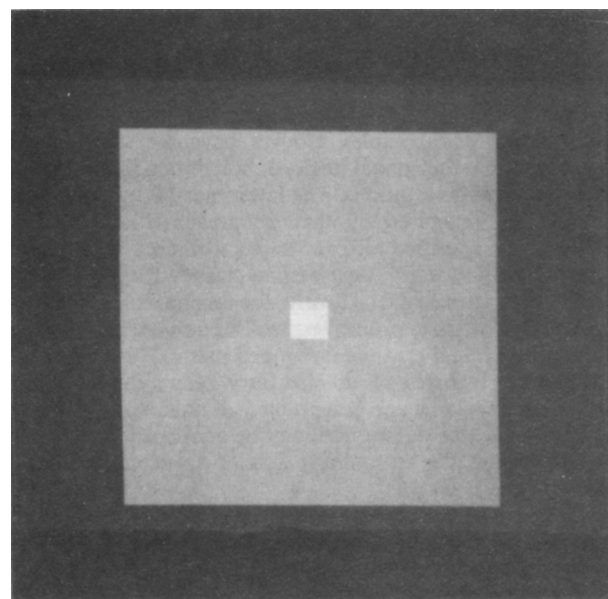


Fig. 1. 50×50 pixel object on dark 512×512 pixel background.

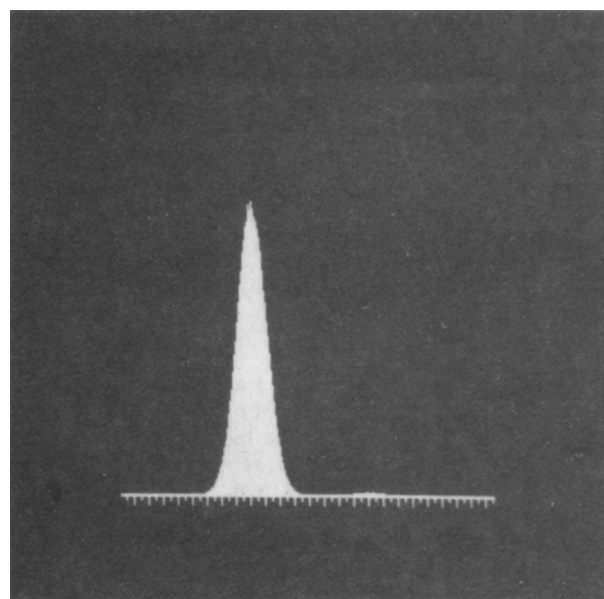


Fig. 2. Gray-level histogram of Fig. 1.

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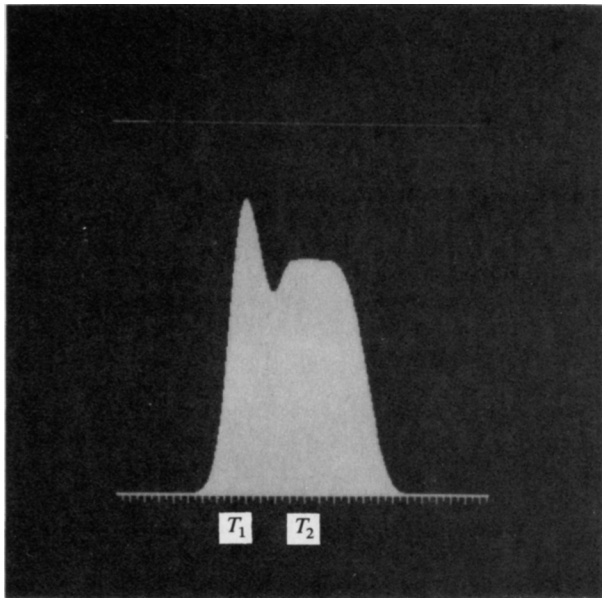
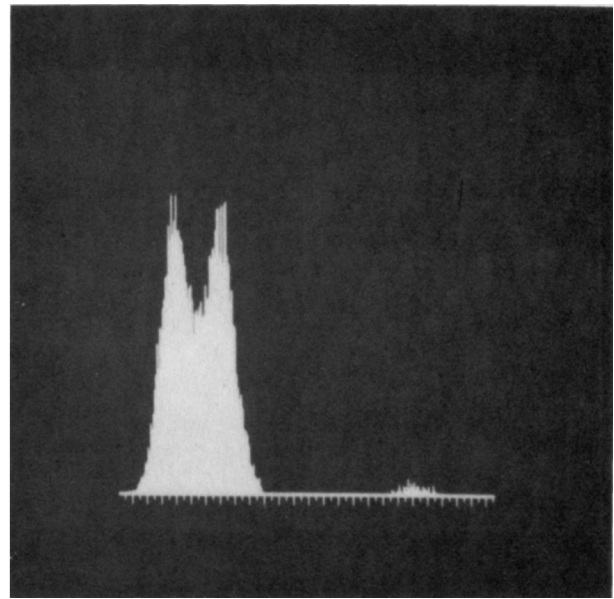
Fig. 3. Criterion function $J_0(T)$ corresponding to image of Fig. 1.

Fig. 4. Trimodal gray-level histogram.

simple task of finding the global maximum. Instead, we need to analyze the function in detail to determine all local peaks T_i . As the function is likely to be considerably smoother than the histogram, there may still be merit in selecting the optimal threshold via $J_0(T)$. However, the need for analyzing this function seriously compromises the assumed competitive edge of Otsu's method over histogram analysis techniques [4].

Second, once the local peaks are determined, we require an additional criterion to decide which peak is associated with good threshold. A simple and effective method is to compare the gray level histogram value at the candidate threshold point T_i with the histogram values at the means μ_{i1} and μ_{i2} of the two populations formed by dividing the histogram at threshold level T_i . Denoting the histogram of gray levels g by $h(g)$, the candidate threshold T_i should satisfy

$$h(T_i) < h(\mu_{i1}) \quad (2)$$

and

$$h(T_i) < h(\mu_{i2}). \quad (3)$$

Note that the use of this condition will correctly lead to the rejection of threshold T_1 in Fig. 3 in favor of cutoff point T_2 .

If the histogram is noisy, it may be more reliable to average the histogram values in the neighborhood of T_i , μ_{i1} , and μ_{i2} . More specifically, conditions (2) and (3) could be replaced by, for instance

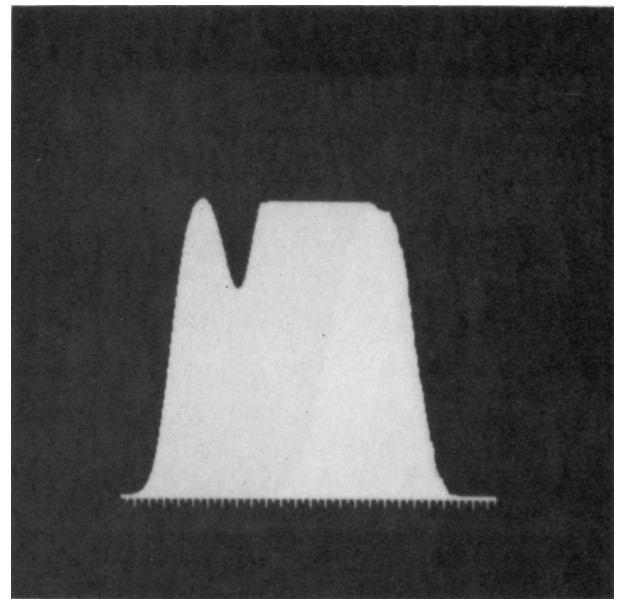
$$\sum_{j=-1}^1 h(T_1 + j) < \sum_{j=-1}^1 h(\mu_{i1} + j) \quad (4)$$

and

$$\sum_{j=-1}^1 h(T_i + j) < \sum_{j=-1}^1 h(\mu_{i2} + j). \quad (5)$$

If both candidate thresholds satisfy (2) and (3), then this implies that both thresholds are likely to be equally valid. In other words, the image cannot be binarized without losing meaningful information. An example of image histogram illustrating such a situation is given in Fig. 4 with the associated Otsu's objective function shown in Fig. 5. The proper action should be to represent the gray-level image by three distinct levels.

Conditions (2) and (3), which we shall refer to as the "valley check," are very useful even if the objective function has only one

Fig. 5. $J_0(T)$ for histogram in Fig. 4.

mode, as this may arise under a number of semantically different situations. First of all, for images with two dominant gray levels (i.e., a bimodal histogram) the valley check will be satisfied, thus confirming that binarization is meaningful. However, if none of the conditions are satisfied, this implies that the histogram is either unimodal, summarizing the gray level statistics of a homogeneous image, or bimodal, with the relative population size of one of the modes exceedingly large. Figs. 6 and 7 show the histogram and the objective function representing the former case. An example of the latter case would be if the size of the bright square in Fig. 1 was reduced to 15×15 pixels. As these two cases give the same result, a more sophisticated method would be needed to determine a good threshold.

Finally, Figs. 8 and 9 illustrate the case of another class of images where the statistics of two distinct populations are such that the gray-level histogram is unimodal. A typical example is the magnitude output of edge detection operation (i.e., a raw edge map), where the class of true edges frequently fails to give rise to a distinct mode in the histogram of edge magnitudes. Fig. 8 is the

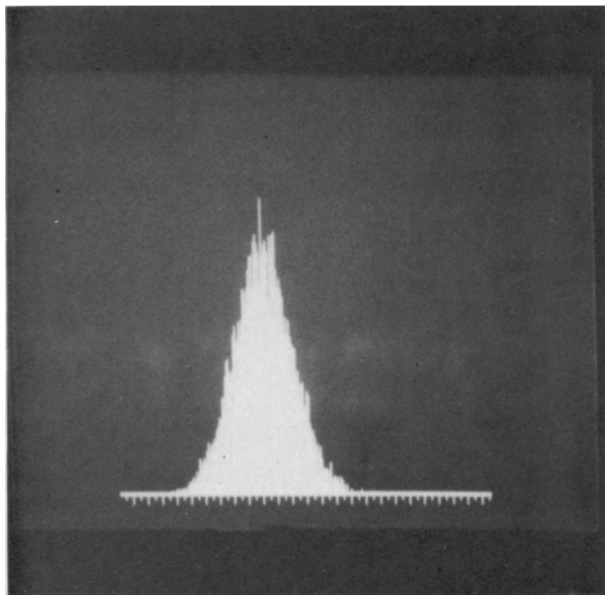


Fig. 6. Unimodal histogram corresponding to noisy homogeneous image.

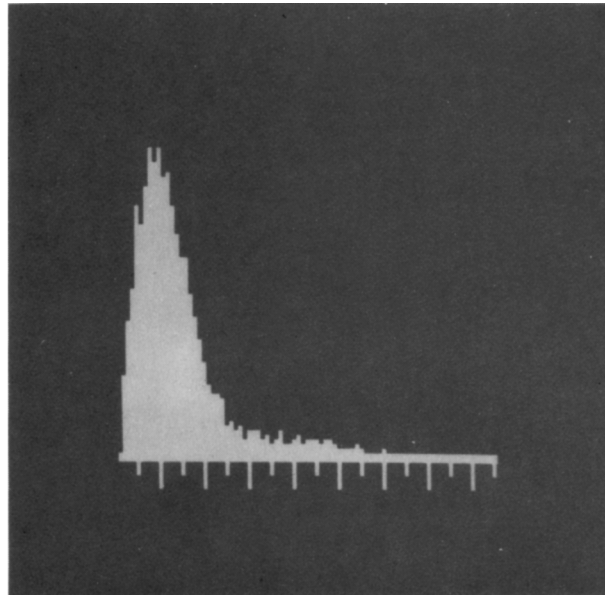
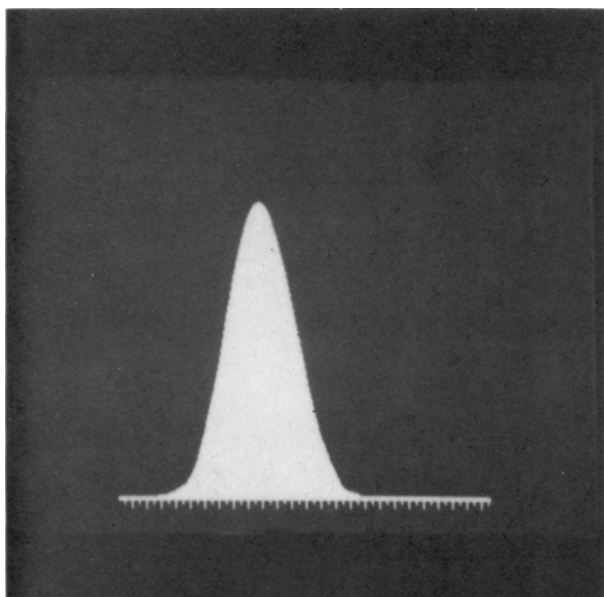
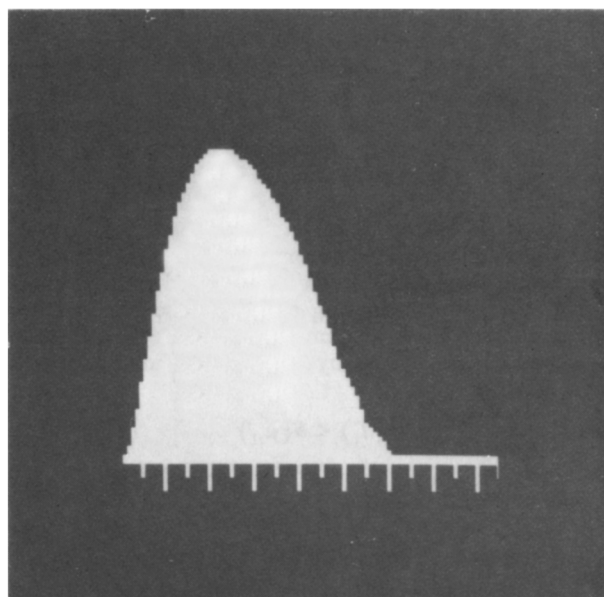


Fig. 8. Unimodal histogram of edge magnitude values.

Fig. 7. $J_0(T)$ for histogram in Fig. 6.Fig. 9. $J_0(T)$ for histogram in Fig. 8.

histogram of edge magnitudes of an image containing a circular object. This situation will also be characterized by a unimodal objective function as shown in Fig. 9. However, here only one of the valley check conditions will be satisfied, thus flagging that the image may be thresholdable. Note that the point of maximum of the objective function will not necessarily be a good threshold. Additional information will be required to select a good threshold value. For instance, in the case of magnitudes, gray level thinning [7] will enhance the separability of edge-nonedge modes in the histogram and, it is hoped, convert it into the type of bimodal histogram shown in Fig. 1.

COMMENTS OF ITERATIVE IMPLEMENTATION

It can be easily shown, e.g., [2], [5] that the criterion in (1) is related to the total variance of gray levels g in the image by

$$\text{var } g = J_v(T) + J_0(T) \quad (6)$$

where $J_v(T)$ is the average variance in the two populations. Since the total variance $\text{var } g$ in the data is constant, instead of maxi-

mizing the between population separation embodied in $J_0(T)$, one could minimize the within population variance $J_v(T)$ and obtain the same optimal threshold. This possibility was considered by Otsu but later dismissed as it was believed that minimization of variance would be more involved because of the need to compute second-order statistical moments. In fact, criterion $J_v(T)$ can be minimized using the iterative Isodata algorithm [5], which also requires only the computation of the means of the two populations and not the variances. After selecting an arbitrary initial threshold, this algorithm computes the means of the resulting populations. The updated threshold is then obtained as the midpoint between the two means. The algorithm terminates when the threshold becomes stable. This algorithm has been used by Ridler [3] directly on the pixel gray-level data and by Trussell [6] on a statistical summary of the data in the form of the histogram. The threshold selection methods [2], [3], [6] should yield similar or identical results.

The comment made earlier concerning the behavior of the criterion function $J_0(T)$ is of course also relevant to Ridler's

method [3] and its more efficient implementation [6]. Its implication is that the minimization algorithm may converge to a local minimum which does not correspond to a good threshold value. Here, however, the remedy is relatively simple. Note that the point of convergence will depend on the initial partition of the grey level range. Suppose we successively apply the threshold selection algorithm for two initial thresholds sufficiently distant from each other. If the two results yielded by the algorithm are identical, we know that criterion function $J_0(T)$ is likely to be unimodal. If the results differ, we have to select the best threshold level from the two candidates. In either case, since we know the means of the two populations, the valley check can easily be used to assess each selected threshold.

CONCLUSION

The incorrectness of Otsu's conjecture of unimodality for the clustering criterion $J_0(T)$, has been demonstrated. We have shown that a simple valley check can be used to enable the threshold selection method to be extended to estimate thresholds reliably over a larger range of population size ratios of object and background pixels. We have also discussed how this problem relates to the dynamic clustering method of threshold selection suggested by Ridler.

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Visual Undersampling in Raster Sampled Images

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Abstract—When a periodic signal is sampled at a rate below the Nyquist limit, it is considered undersampled because it is impossible to recover the original signal from the samples without some additional signal information. We have studied the visual effects of raster sampling upon the discriminability of simple monochrome test targets (edges and sinusoidal gratings) and found out that observers have good perceptual tolerance to the undersampling of wideband spatial edges. However, for sinusoidal test gratings the low sampling rates produce illusory visual effects that disturb the recognition of the waveform. The occurrence of these illusions in sampled images points out two different sources of sampling noise in images: genuine spectral noise and purely visual spatial noise. Our findings suggest that scenes containing abundant edge information tolerate low sampling rates better than scenes in which low and medium spatial frequencies are dominant.

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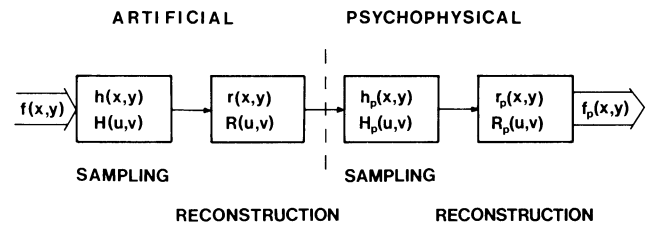


Fig. 1. Image sampling and reconstruction in artificial systems and human spatial vision. Spatial sampling and reconstruction of an image $f(x,y)$ corresponds to filtering by $H(u,v)$ and $R(u,v)$.

INTRODUCTION

When an image is sampled for digital or other signal processing purposes, its sampled representation will contain both the original signal information and sampling noise. According to the sampling theorem, reconstruction of the image from its samples is possible by interpolation between the sample points provided that the sampling has been sufficiently dense [1]–[3]. Reconstruction filtering can be accomplished by proper optical, analog, or digital filters. When the human observer is the final link in the image processing chain, his visual system forms an additional reconstruction stage in this cascade of filters. However, it is not known how effective the visual system is in reconstructing sampled scenes.

For physical filters the reconstruction capacity can be estimated by determining how much reconstruction noise or aliased frequency components is introduced to the reconstructed signal [4]. To measure the corresponding performance for the human visual system, we have approached this problem within the general framework described in Fig. 1. Fig. 1 shows that the visual system can be divided into two filtering stages, which determine the properties of the perceived image $f_p(x,y)$. These stages consist of the visual sampling and reconstruction filters $H_p(u,v)$ and $R_p(u,v)$. In foveal vision, which is predominantly used in normal viewing, the final quality of the perceived image is determined mainly by the visual reconstruction filter $R_p(u,v)$ because the visual sampling processes in the fovea are of a relatively high quality.

Our aim is to develop psychophysical measures that would be suitable for describing the visual system's capacity to reconstruct different types of image signal from their sampled versions. Only δ -sampling will be considered here; we have shown earlier that δ -sampling can be used to reveal the spatial interpolation that the visual system performs to sampled images [5].

The visual reconstruction capacity can be estimated only by appropriate psychophysical measurements of the observer performance in tasks that require him to recognize a sampled scene. A simple measure of the observer's visual reconstruction capacity is given by the minimum sampling rate that is required for the recognition of a sinusoidal test grating. It would be expected to be systematically dependent on the spatial frequency of the grating. On the other hand, if we want to determine how well the visual system tolerates sampling noise, a threshold measure is needed that describes the occurrence of the aliasing type of visual noise in the sampled image.

To study the visual system's capacity to reconstruct simple test targets which have been sampled at low rates, we have used spatial edges and sinusoidal gratings as test stimuli. Their line sampled versions have been presented to the observers to determine two different thresholds: the threshold sampling rate for the recognition of the edge waveform and the threshold for the occurrence of the visual undersampling effects. It will be shown that these measures are not necessarily directly related to the physical sampling constraints. For example, a physically mild